

ON THE DEFINITION AND SPECIFICATION OF STATIONARY POINT PROCESSES

V.W. Steinijs
University of Natal, Durban

SUMMARY

In this study the measure theoretic concepts of Ryll-Nardzewski (1961) are used to give a rigorous definition of a stationary point process and some related processes. Particular consideration is given to the lengths of time intervals between successive events of a realisation. This leads to two different types of random variables depending on "whether the origin of observation is an arbitrary time instant or an arbitrary event", as e.g. McFadden (1962) phrases it, and I shall speak about lengths and distances, respectively. It is shown how various results relating the distributions of lengths and distances, which so far have only been derived heuristically, follow straight away from the results of Ryll-Nardzewski (1961).

Furthermore, it is shown how the basic probability space can be uniquely specified by the number of events in intervals (see section 2) as well as by the finite dimensional distribution functions of the distances (see section 4).

1. Definition of a stationary point process

Let Ω denote the class of all countable subsets of the real line $\mathbb{R}$ that are finite in each finite interval. $\omega \in \Omega$ may be considered as a sequence of events occurring at the instants

$$T_i(\omega), i \in \mathbb{I} := \{0, \pm 1, \pm 2, \dots\},$$

where the indices are allocated such that for $i, j \in \mathbb{I}$ with $i < j$ $T_i(\omega) < T_j(\omega)$, $T_0(\omega)$ denoting the largest non positive value in the sequence.

Let $\mathbf{B}$ denote the Borel algebra over $\mathbb{R}$, and let $\bar{\mathbf{B}}$ denote the extended Borel algebra over the extended real numbers $\bar{\mathbb{R}}$.

Consider the functions

$$N(., B) : \Omega \rightarrow \bar{\mathbb{R}},$$

defined for $B \in \mathbf{B}$ by

$$N(\omega, B) := \text{card}(\omega \cap B),$$

and the σ -algebra induced by them, i.e.

$$(1) \quad \mathbf{A} := \sigma\left(\bigcup_{B \in \mathbf{B}} N(., B)^{-1} \bar{\mathbf{B}}\right).$$

If a probability measure (henceforth abbreviated as p -measure) $P|_{\mathbf{A}}$ is defined, then we call the family of random variables $(N(., B))_{B \in \mathbf{B}}$ the point process defined on the p -space $(\Omega, \mathbf{A}, P)$. In order to introduce stationarity let for $t \in \mathbb{R}$ and $A \in \mathbf{A}$

$$\omega \in A^t \text{ iff } \omega - t \in A,$$

where

$$\omega - t = (T_i(\omega) - t)_{i \in \mathbb{I}} \text{ if } \omega = (T_i(\omega))_{i \in \mathbb{I}}.$$

Similarly let for $B \in \mathbf{B}$ and $r \in \mathbb{R}$

$r \in B^t$ iff $r \cdot t \in B$

Then $P|A$ is called stationary iff

(2) $P(A) = P(A^t)$ for all $t \in \mathbb{R}$ and all $A \in \mathcal{A}$.

The point process $(N(\cdot, B))_{B \in \mathcal{B}}$ is called stationary iff for all $n \in \mathbb{N}$ ($\mathbb{N}$ denoting the set of natural numbers), all $B_i \in \mathcal{B}$, $i=1, \dots, n$ and all $t \in \mathbb{R}$ the two sets of random variables

$$N(\cdot, B_1), \dots, N(\cdot, B_n); \\ N(\cdot, B_1^t), \dots, N(\cdot, B_n^t)$$

have the same joint distribution (see e.g. Cox and Miller (1968), p. 339-340). Choosing

$$\mathcal{A} = \{\omega \in \Omega : N(\omega, B_1) \leq x_1, \dots, N(\omega, B_n) \leq x_n\}$$

in (2) shows that the stationarity of $P|A$ implies the invariance under time shifts of the finite dimensional distribution functions of the $N(\cdot, B_i)$, $i=1, \dots, n$, and hence the stationarity of the point process. Furthermore, as $\mathcal{A}$ is the σ -algebra induced by $(N(\cdot, B))_{B \in \mathcal{B}}$, the converse relation is easily shown, and hence stationarity of $P|A$ and stationarity of the point process $(N(\cdot, B))_{B \in \mathcal{B}}$ are equivalent.

Closely related to a stationary point process is the so-called counting process

$$N(\omega, (t, t+\tau]) \quad (t, \tau) \in \mathbb{R} \times (0, \infty),$$

and we only remark that the stationarity of $P|A$ implies the stationarity of the counting process with respect to the parameter t for fixed τ .

2. Specification of a p-measure on $(\Omega, \mathcal{A})$ through the distribution of numbers of events in intervals

Let $N_0 := N \cup \{0\}$, and let $\mathbf{J}$ denote the class of all finite, left open and right closed intervals in $\mathbb{R}$. Let a set

$$\{\omega \in \Omega : a_i \leq N(\omega, J_i) < b_i : i = 1, \dots, n\},$$

where $a_i \geq -1$, $b_i \geq 0$, $J_i \in \mathbf{J}$, $i=1, \dots, n$, $n \in \mathbb{N}$, and where the sign $\leq$ may be replaced by $<$ and vice versa (the sign next to $+\infty$ should always be $<$), be called a cylinder set with n -dimensional base interval. Then for each $n \in \mathbb{N}$ and all $m_1, \dots, m_n \in N_0$ and all pairwise disjoint $J_1, \dots, J_n \in \mathbf{J}$ postulate

$$P_{J_1, \dots, J_n}(m_1, \dots, m_n)$$

to be the probability of the set

$$\{\omega \in \Omega : N(\omega, J_1) = m_1, \dots, N(\omega, J_n) = m_n\},$$

such that the following conditions are met:

$$(3) \quad P_{J_1, \dots, J_n}(m_1, \dots, m_n) \geq 0;$$

$$(4) \quad \sum_{n \in N_0} P_J(n) = 1 \quad \text{for all } J \in \mathbf{J};$$

$$(5) \quad P_{J_1, \dots, J_n}(m_1, \dots, m_n) \\ = \sum_{\substack{n^{(1)}, n^{(2)} \in N_0 \\ n^{(1)} + n^{(2)} = m_n}} P_{J_1, \dots, J_{n-1}, J^{(1)}, J^{(2)}}(m_1, \dots, m_{n-1}, n^{(1)}, n^{(2)});$$

$$\begin{aligned}
 & \text{for all } J^{(1)}, J^{(2)} \in \mathbf{J} \text{ with } J_n = J^{(1)} + J^{(2)} \text{ (the } + \text{ sign indicating } J^{(1)} \cap J^{(2)} = \emptyset); \\
 (6) \quad & P_{J_1, \dots, J_n} (m_1, \dots, m_n) = P_{J_{i_1}, \dots, J_{i_n}} (m_{i_1}, \dots, m_{i_n}) \\
 & \text{for all permutations } (i_1, \dots, i_n) \text{ of } (1, \dots, n) \text{ and finally} \\
 (7) \quad & \sum_{k \in \mathbb{N}_0} P_{J_1, \dots, J_n} (m_1, \dots, m_{n-1}, k) = P_{J_1, \dots, J_{n-1}} (m_1, \dots, m_{n-1})
 \end{aligned}$$

Whereas conditions (3) and (4) obviously must be met, (5) enables the unique allocation of a probability value to any finite collection of not necessarily disjoint intervals in $\mathbf{J}$. Finally the symmetry condition (6) and the consistency condition (7) are needed to define the probability of a cylinder set in a unique way (see Nawrotzki (1962), p. 202-205, to whom the specification given so far is due).

By finite additivity this probability P is extended to all finite sums of cylinder sets which form an algebra $\mathbf{A}^*$. Obviously $P|_{\mathbf{A}^*}$ is finite. Hence to each cylinder set C with n -dimensional base interval there exists a subset K of C with closed and bounded base interval in $\mathbb{R}^n$ such that $P(C \setminus K)$ can be made arbitrarily small. From this it can be shown that $P|_{\mathbf{A}^*}$ is σ -additive (see e.g. Cramér and Leadbetter (1967), p. 34-36), so that $P|_{\mathbf{A}^*}$ can be uniquely extended to a p -measure $P|_{\sigma\mathbf{a}(\mathbf{A}^*)}$ on the σ -algebra generated by $\mathbf{A}^*$.

We now show $\sigma\mathbf{a}(\mathbf{A}^*) = \mathbf{A}$. As $N(\omega, B)$, $B \in \mathbf{B}$, is $\mathbf{A}, \mathbf{B}$ -measurable, it follows $\sigma\mathbf{a}(\mathbf{A}^*) \subset \mathbf{A}$. To see the converse relation observe that the family of all Borel sets in definition (1) can be replaced by the family $\mathbf{I}$ of all intervals of the real line, i.e.

$$\mathbf{A} = \sigma\mathbf{a}\left(\bigcup_{J \in \mathbf{I}} N(\cdot, J)^{-1} B\right).$$

Now as $\omega \in \Omega$ cannot have a finite limit point, each inverse image under $N(\cdot, J)$, J finite or infinite, can be written in terms of countable unions and intersections of sets in $\mathbf{A}^*$. Hence $\mathbf{A} \subset \sigma\mathbf{a}(\mathbf{A}^*)$, so that $P|_{\mathbf{A}^*}$ has been uniquely extended to a p -measure $P|_{\mathbf{A}}$.

$P|_{\mathbf{A}}$ thus specified becomes stationary iff in addition to (3)-(7) it is demanded that for all $t \in \mathbb{R}$, all $n \in \mathbb{N}$, all $m_i \in \mathbb{N}_0$ and all pairwise disjoint $J_i \in \mathbf{J}$ ($i=1, \dots, n$).

$$(8) \quad P_{J_1^t, \dots, J_n^t} (m_1, \dots, m_n) = P_{J_1, \dots, J_n} (m_1, \dots, m_n).$$

This follows as (8) implies the invariance of $P|_{\mathbf{A}^*}$ under time shifts, and as the extension from $P|_{\mathbf{A}^*}$ to $P|_{\mathbf{A}}$ is unique.

As stationarity of $P|_{\mathbf{A}}$ implies

$$P\left\{\omega \in \Omega : N(\omega, \mathbb{R}) = 0 \text{ or } N(\omega, (-\infty, 0)) = N(\omega, (0, \infty)) = +\infty\right\} = 1,$$

we can restrict ourselves to the non trivial case that there are infinitely many events in both time directions, if we assume that $P\{\omega \in \Omega : N(\omega, \mathbb{R}) = 0\} = 0$. From that it follows in particular that the point process $(N(\omega, B))_{B \in \mathbf{B}}$ has a strictly positive intensity

$$\nu := \int N(\omega, (0, 1]) P(d\omega),$$

a result to be referred to later on.

3. The processes of distances and lengths.

Let $(\Omega, \mathbf{A}, P)$ be specified. Consider the lengths of the intervals between successive events of $\omega = (T_i(\omega))_{i \in \mathbb{I}}$:

$$L_i(\omega) := T_{i+1}(\omega) - T_i(\omega), i \in I.$$

As e.g. for $i \geq 1$

$$T_i(\omega) < b \text{ iff } N(\omega, (0, b)) \geq i,$$

the $T_i(\omega)$, $i \in I$, are $\mathbf{A}, \mathbf{B}$ -measurable, and so are the $L_i(\omega)$, $i \in I$, as differences of measurable functions. We shall call the $L_i(\omega)$ considered as random variables on the p -space $(\Omega, \mathbf{A}, P)$ the lengths, and the family $(L_i(\cdot))_{i \in I}$ the process of lengths.

Consider now the subspace of all sequences that contain a particular event, say at $t=0$, i.e.

$$\Omega_0 := \{\omega \in \Omega : \omega \wedge \{0\} \neq \emptyset\},$$

and the corresponding σ -algebra

$$\mathbf{A}_0 := \mathbf{A} \cap \Omega_0.$$

If now $P_0 | \mathbf{A}_0$ is a p -measure which can be considered as conditional probability given an event at $t=0$, then we call the $\mathbf{A}_0, \mathbf{B}$ -measurable lengths considered as random variables on $(\Omega_0, \mathbf{A}_0, P_0)$ the distances. We denote the distances by $X_i(\omega)$, $i \in I$, instead of $L_i(\omega)$, $i \in I$, and call the family $(X_i(\cdot))_{i \in I}$ of random variables on $(\Omega_0, \mathbf{A}_0, P_0)$ the process of distances.

Obviously this definition describes precisely what e.g. McFadden (1962) and Cox and Lewis (1966) mean when they speak about "two sets of random lengths depending on whether one starts with an arbitrary instant or with an arbitrary event".

We are now interested in such a p -measure $P_0 | \mathbf{A}_0$ with respect to which in addition the distances form a stationary sequence, i.e. for all $n \in \mathbb{N}$, all $j \in I$ and all $i_1, \dots, i_n \in I$ the two sets of random variables on $(\Omega_0, \mathbf{A}_0, P_0)$

$$X_{i_1}(\omega), \dots, X_{i_n}(\omega)$$

$$X_{i_1+j}(\omega), \dots, X_{i_n+j}(\omega)$$

have the same joint distribution, which is equivalent to the invariance of $P_0 | \mathbf{A}_0$ under time shifts by the sums $T_i(\omega)$, $i \in I$, of consecutive distances:

$$P_0(A_0) = P_0(A_0 \cap T_i(\omega))$$

for all $i \in I$ and all $A_0 \in \mathbf{A}_0$.

Ryll-Nardzewski (1961) gives a rather elegant derivation of a p -measure $P_0 | \mathbf{A}_0$ as postulated above. For a stationary point process with finite intensity ν he considers the σ -finite measures $\mu_A | \mathbf{B}$ ($A \in \mathbf{A}$) defined by

$$\mu_A(\mathbf{B}) := \int \chi_A(\omega) N(\omega, \mathbf{B}) P(d\omega), \mathbf{B} \in \mathbf{B},$$

and $\mu | \mathbf{B}$ defined by

$$\mu(\mathbf{B}) := \int N(\omega, \mathbf{B}) P(d\omega), \mathbf{B} \in \mathbf{B},$$

and shows the existence of Radon-Nikodym derivatives

$$P(A | t) := \frac{d\mu_A}{d\mu} : \mathbf{A} \times \mathbb{R} \rightarrow [0, 1]$$

which are such that $P(\cdot|t)|A$ is a p-measure for each $t \in R$. Furthermore in the equivalence class of these functions there exists a unique version for which in addition $P(A|t) = P(A^s|t+s)$ for all $A \in \mathcal{A}$ and all $s, t \in R$. Furthermore e.g. $P_0|A := P(\cdot|0)|A$ has the following properties which justify to call it a conditional probability given an event at $t=0$:

$$\lim_{|I_0| \rightarrow 0} E_P [f | N(\omega, I_0) \geq 1] = E_{P_0} [f]$$

for each bounded $f: \Omega \rightarrow R$ such that $f(\omega+t)$ for fixed $\omega \in \Omega$ considered as a function of $t \in R$ is continuous, where I_0 denotes an interval of length $|I_0|$, which contains the origin;

$$P_0(\Omega_0) = 1$$

(see Ryll-Nardzewski (1961), p. 462).

The general relation between $P|A$ and $P_0|A_0$ is given by the following theorem (see Ryll-Nardzewski (1961), p. 463-465. Theorem 7): The distances $X_i(\omega)$, $i \in I$, on the p-space (Ω_0, A_0, P_0)

- (i) are positive
- (ii) have finite first moment

$$E_{P_0} [X_i] = \frac{1}{\nu}, i \in I,$$

ν being the positive intensity of

$$(N(\cdot, B))_{B \in \mathcal{B}}$$

- (iii) form a stationary sequence.

Conversely to each sequence $X_i(\omega)$, $i \in I$, of random variables on (Ω_0, A_0, P_0) satisfying (i)-(iii) there exists exactly one p-measure $P|A$ that is stationary and such that $P_0|A_0$ coincides with the conditional probability given 0 as derived above.

Moreover, the correspondence between $P|A$ and $P_0|A_0$ is one to one and is given by

$$(9) \quad \int f(\omega) P(d\omega) = \nu \int_{\Omega_0} P_0(d\omega) \int_0^{X_0(\omega)} f(\omega-t) dt$$

for all P-integrable functions $f| \Omega$.

Let us now see how various results connecting the distributions of the lengths and the distances follow straight away from (9).

First of all, stationarity implies that all $X_i(\omega)$, $i \in I$, induce the same p-measure $Q_X|B$, defined by

$$Q_X(B) := P_0 \{ \omega \in \Omega_0 : X_0(\omega) \in B \}, B \in \mathcal{B}.$$

The lengths $L_i(\omega)$, $i \in I$, however, will for different $i \in I$ be differently distributed. Let $Q_{L_0}|B$ be the p-measure induced by the length $L_0(\omega)$ of the interval in progress at $t=0$:

$$Q_{L_0}(B) := P \{ \omega \in \Omega : L_0(\omega) \in B \}, B \in \mathcal{B}.$$

Paying attention to the fact of getting a length biased $L_0(\omega)$ if one starts observing without knowledge of the particular realisation $\omega \in \Omega$, a heuristic argument leads to the following relation between $Q_{L_0}|B$ and $Q_X|B$:

$$(10) \quad \int_0^x Q_{L_0}(dt) = \nu \int_0^x t Q_X(dt), x \geq 0$$

(see e.g. McFadden (1962), p. 366).

By setting $f(\omega) = \chi_{\{\omega \in \Omega : L_0(\omega) \leq x\}}(\omega)$, $x \geq 0$, the l.h.s. of (9) yields

$$\begin{aligned} \int f(\omega) P(d\omega) &= P\{\omega \in \Omega : L_0(\omega) \leq x\} \\ &= Q_{L_0}\{t \in \mathbb{R} : t \leq x\} \\ &= \int_0^x Q_{L_0}(dt), \end{aligned}$$

and hence the l.h.s. of (10).

To evaluate the r.h.s. of (9) we observe that for $\omega \in \Omega$ $X_0(\omega) = L_0(\omega)$ and for $t \in [0, X_0(\omega)]$ $L_0(\omega - t) = L_0(\omega)$.

Hence

$$\begin{aligned} & \nu \int_{\Omega_0} P_0(d\omega) \int_0^{X_0(\omega)} \chi_{\{\omega \in \Omega : L_0(\omega) \leq x\}}(\omega - t) dt \\ &= \nu \int_{\Omega_0} X_0(\omega) \chi_{\{\omega \in \Omega_0 : X_0(\omega) \leq x\}}(\omega) P_0(d\omega) \\ &= \nu \int_0^x t Q_X(dt), \end{aligned}$$

which is the r.h.s. of (10). It should be mentioned that whenever $f(\omega)$ is measurable with respect to the σ -algebra $\mathcal{L}$ induced by the lengths, the integrand $f(\omega - t)$ in (9) is constant for $t \in [0, X_0(\omega)]$.

Other relations given by McFadden (1962) as e.g. that between the joint distribution of $L_0(\omega)$ and $\sum_{i=1}^n L_i(\omega)$ and the joint distribution of $X_0(\omega)$ and $\sum_{i=1}^n X_i(\omega)$, follow similarly from (9) by choosing a particular $f(\omega)$, in this case the indicator function of

$$\{\omega \in \Omega : L_0(\omega) \leq x, \sum_{i=1}^n L_i(\omega) \leq y\}.$$

Also the moments of the lengths in terms of the moments of the distances are obtainable from (9), e.g. for $i \in \mathbb{I}$ and $k \geq 1$

$$E_P[L_i^k] = \nu E_{P_0}[X_0 X_i^k].$$

4. Specification of a p-measure on $(\Omega, \mathcal{A})$ via the finite dimensional distribution functions of the distances.

Let the finite dimensional distribution functions $F(x_1, \dots, x_n; i_1, \dots, i_n)$ of the distances $X_{i_1}, \dots, X_{i_n}$ ($i_j \in \mathbb{I}$, $x_j \in \mathbb{R}$ ($j = 1, \dots, n$), $n \in \mathbb{N}$) fulfil the symmetry condition

$$F(x_{p_1}, \dots, x_{p_n}; i_{p_1}, \dots, i_{p_n}) = F(x_1, \dots, x_n; i_1, \dots, i_n)$$

for all permutations $(p_1, \dots, p_n)$ of $(1, \dots, n)$ and the consistency condition

$$\lim_{x_n \rightarrow \infty} F(x_1, \dots, x_n; i_1, \dots, i_n) = F(x_1, \dots, x_{n-1}; i_1, \dots, i_{n-1})$$

Then by Kolmogorov's theorem (see e.g. Cramér and Leadbetter (1967), p. 34-37) a unique p-measure is defined on the Borel-algebra $\mathbf{B}^I$ over the space R^I of I-tuples of real numbers. As $\mathbf{B}^I$ is the σ -algebra generated by the cylinder sets

$$\{ a_j < X_{i_j} < b_j : j = 1, \dots, n, i_j \in I, a_j, b_j \in R, n \in \mathbb{N} \}$$

this p-measure obviously induces a p-measure on the σ -algebra induced by the distances, say $\bar{P}_0|D$, where

$$D := \sigma\left(\bigcup_{i \in I} X_i(\cdot)^{-1}B\right).$$

We shall show $D = A_0$ and shall then use (9) to extend $\bar{P}_0|A_0$ to a p-measure $\bar{P}|A$. Obviously $D \subset A_0$, as the distances are A_0, B -measurable. Conversely to see $A_0 \subset D$ let $\omega \in \Omega_0$ and $(a, b]$ any real interval, without loss of generality finite. Then as for $n \in \mathbb{N}$

$$\{\omega \in \Omega_0 : N(\omega, (a, b]) \geq n\} = \bigcup_{k \in I} \{\omega \in \Omega_0 : T_{k+1}(\omega) > a, T_k + n(\omega) \leq b\}$$

and as

$$\begin{aligned} T_i(\omega) &= \sum_{j=0}^{i-1} X_j(\omega) & \text{for } i \geq 1, \\ T_i(\omega) &= -\sum_{j=i}^{-1} X_j(\omega) & \text{for } i < 0, \end{aligned}$$

it follows that $N(\omega, J)|\Omega_0$ is $D, \bar{B}$ -measurable for $J \in I$ and hence $A_0 \subset D$.

$\bar{P}_0|A_0$ thus uniquely defined is stationary with respect to the process of distances if the finite dimensional distribution functions of the distances in addition to symmetry and consistency are invariant under index-shifts, i.e. if for $j \in I$ and $i_1, \dots, i_n \in I$

$$F(x_1, \dots, x_n; i_1 + j, \dots, i_n + j) = F(x_1, \dots, x_n; i_1, \dots, i_n).$$

If finally the distances have finite first moment with respect to $\bar{P}_0|A_0$, then there exists – as pointed out in section 3 – exactly one stationary p-measure $\bar{P}|A$ such that $\bar{P}_0|A_0$ can be considered as conditional probability given an event at $t=0$. Furthermore from (9) we get

$$\bar{P}(A) = \nu \int_{\Omega_0} \bar{P}_0(d\omega) \int_0^{X_0(\omega)} \chi_{A_t(\omega)} dt, A \in A,$$

which for sets A in the σ -algebra L induced by the lengths reduces to

$$\bar{P}(A) = \nu \int X_0(\omega) \chi_{A \cap \Omega_0}(\omega) \bar{P}_0(d\omega),$$

as $A \in L$ implies

$$\chi_{A_t}(\omega) = \chi_A(\omega) \text{ for } t \in [0, X_0(\omega)).$$

Acknowledgement

I wish to thank Professor H. Linhart for encouraging me to write this paper.

REFERENCES

- COX, D.R. and LEWIS, P.A.W. (1966). *Statistical analysis of series of events*. Methuen, London.
- COX, D.R. and MILLER, H.D. (1968). *The theory of stochastic processes*. Methuen, London.
- CRAMÉR, H. and LEADBETTER, M.R. (1967). *Stationary and related stochastic processes*. Wiley, New York.
- McFADDEN, J.A. (1962). On the lengths of intervals in a stationary point process. *J. Roy. Stat. Soc. B.* **24**, 364-382.
- NAWROTZKI, K. (1962). Ein Grenzwertsatz für homogene zufällige Punktfolgen. *Math. Nachr.* **24**, 201-217.
- RYLL-NARDZEWSKI, C. (1961). Remarks on processes of calls. *Proc. 4th Berkeley Symp.* **2**, 455-465.

Manuscript received 9th June 1969.